and $(a+(-b)\sqrt{n}) \in \mathbb{Z}[\sqrt{n}]$, so α is a unit.

c. The preceding parts show that we have a multiplicative norm on $\mathbb{Z}[\sqrt{n}]$ such that the elements of norm 1 are precisely the units. By Exercise 13, every nonzero nonunit has a factorization into irreducibles.

18. Given α and β in $\mathbb{Z}[\sqrt{-2}]$, we proceed to construct $\sigma = q_1 + q_2\sqrt{-2}$ and $\rho = \alpha - \beta\sigma$ as described in the proof of Theorem 47.4. Viewing $\alpha = a + b\sqrt{-2} = a + (b\sqrt{2})i$ in $\mathbb{C}$, we have $|a + (b\sqrt{2})i|^2 = a^2 + 2b^2 = N(\alpha)$. Working in $\mathbb{C}$, with $|\ |^2$, we compute

$$\frac{\alpha}{\beta} = \frac{a+b\sqrt{-2}}{c+d\sqrt{-2}} \cdot \frac{c-d\sqrt{-2}}{c-d\sqrt{-2}} = r + s\sqrt{-2}$$

for $r, s \in \mathbb{Q}$. Again, we choose q_1 and q_2 to be integers in $\mathbb{Z}$ as close as possible to r and s respectively.

By construction of σ, we see that $|r - q_1| \le 1/2$ and $|s - q_2| \le 1/2$. Thus

$$\begin{aligned} N(\frac{\alpha}{\beta} - \sigma) &= N((r + s\sqrt{-2}) - (q_1 + q_2\sqrt{-2})) \\ &= N((r-q_1) + (s-q_2)\sqrt{-2}) \\ &\le \left(\frac{1}{2}\right)^2 + 2\left(\frac{1}{2}\right)^2 \le \frac{1}{4} + 2 \cdot \frac{1}{4} = \frac{3}{4}. \end{aligned}$$

Thus we obtain

$$N(\rho) = N(\alpha - \beta\sigma) = N(\beta(\frac{\alpha}{\beta} - \sigma)) = N(\beta)N(\frac{\alpha}{\beta} - \sigma) \le N(\beta)\frac{3}{4},$$

so we do indeed have $N(\rho) < N(\beta)$ as claimed.

48. Automorphisms of Fields

1. The conjugates of $\sqrt{2}$ over $\mathbb{Q}$ are $\sqrt{2}$ and $-\sqrt{2}$.

2. Now $\sqrt{2}$ is the only conjugate of $\sqrt{2}$ over $\mathbb{R}$ because $\sqrt{2} \in \mathbb{R}$.

3. The conjugates of $3+\sqrt{2}$ over $\mathbb{Q}$ are $3+\sqrt{2}$ and $3-\sqrt{2}$; they are the zeros of $(x-3)^2-2 = x^2-6x+7$.

4. The conjugates of $\sqrt{2}-\sqrt{3}$ over $\mathbb{Q}$ are

$$\sqrt{2}-\sqrt{3}, \sqrt{2}+\sqrt{3}, -\sqrt{2}-\sqrt{3}, \text{ and } -\sqrt{2}+\sqrt{3}$$

as indicated by Example 48.17.

5. The conjugates of $\sqrt{2}+i$ over $\mathbb{Q}$ are $\sqrt{2}+i, \sqrt{2}-i, -\sqrt{2}+i$, and $-\sqrt{2}-i$. This is clear because $\mathbb{Q}(\sqrt{2}+i) = (\mathbb{Q}(\sqrt{2}))(i)$.

6. The conjugates of $\sqrt{2}+i$ over $\mathbb{R}$ are $\sqrt{2}+i$ and $\sqrt{2}-i$ because $\sqrt{2} \in \mathbb{R}$.

7. The conjugates of $\sqrt{1+\sqrt{2}}$ over $\mathbb{Q}$ are

$$\sqrt{1+\sqrt{2}}, \sqrt{1-\sqrt{2}}, -\sqrt{1+\sqrt{2}}, \text{ and } -\sqrt{1-\sqrt{2}}.$$

This clear because $\mathbb{Q}(\sqrt{1+\sqrt{2}}) = (\mathbb{Q}(\sqrt{2}))(\sqrt{1+\sqrt{2}})$.

8. The conjugates of $\sqrt{1+\sqrt{2}}$ over $\mathbb{Q}(\sqrt{2})$ are just $\sqrt{1+\sqrt{2}}$ and $-\sqrt{1+\sqrt{2}}$ because $\sqrt{2} \in \mathbb{Q}(\sqrt{2})$.

9. $\tau_2(\sqrt{3}) = \sqrt{3}$.

10. $\tau_2(\sqrt{2}+\sqrt{5}) = -\sqrt{2}+\sqrt{5}$.

11. $(\tau_3\tau_2)(\sqrt{2}+3\sqrt{5}) = \tau_3[\tau_2(\sqrt{2}+3\sqrt{5})] = \tau_3(-\sqrt{2}+3\sqrt{5}) = -\sqrt{2}+3\sqrt{5}$.

12. $(\tau_5\tau_3)\left(\frac{\sqrt{2}-3\sqrt{5}}{2\sqrt{3}-\sqrt{2}}\right) = \tau_5\left[\tau_3\left(\frac{\sqrt{2}-3\sqrt{5}}{2\sqrt{3}-\sqrt{2}}\right)\right] = \tau_5\left(\frac{\sqrt{2}-3\sqrt{5}}{-2\sqrt{3}-\sqrt{2}}\right) = \frac{\sqrt{2}+3\sqrt{5}}{-2\sqrt{3}-\sqrt{2}}$

13. Note that $\sqrt{45} = 3\sqrt{5}$. We have

$$\begin{aligned}(\tau_5{}^2\tau_3\tau_2)(\sqrt{2}+\sqrt{45}) &= \tau_5{}^2\tau_3[\tau_2(\sqrt{2}+\sqrt{45})]\\ &= \tau_5{}^2[\tau_3[(-\sqrt{2}+\sqrt{45})]\\ &= \tau_5{}^2(-\sqrt{2}+\sqrt{45})\\ &= \tau_5(-\sqrt{2}-\sqrt{45}) = -\sqrt{2}+\sqrt{45}.\end{aligned}$$

14. Note that $\sqrt{30} = \sqrt{2}\sqrt{3}\sqrt{5}$. Now $\tau_3[\tau_5(\sqrt{2}-\sqrt{3}+(\tau_2\tau_5)(\sqrt{30}))]$

$$\begin{aligned}&= \tau_3[\tau_5(\sqrt{2}-\sqrt{3}+\tau_2(\tau_5(\sqrt{30})))]\\ &= \tau_3[\tau_5(\sqrt{2}-\sqrt{3}+\tau_2(-\sqrt{30}))]\\ &= \tau_3[\tau_5(\sqrt{2}-\sqrt{3}+\sqrt{30})]\\ &= \tau_3(\sqrt{2}-\sqrt{3}-\sqrt{30}) = \sqrt{2}+\sqrt{3}+\sqrt{30}.\end{aligned}$$

15. **a.** Because $\sigma_1(\sqrt{2}) = -\sqrt{2}, \sigma_3(\sqrt{3}) = -\sqrt{3}$, and $\sigma_1(\sqrt{6}) = -\sqrt{6}$, the only elements of $\mathbb{Q}(\sqrt{2},\sqrt{3})$ left fixed by both σ_1 and σ_3 are those in $\mathbb{Q}$, so $\mathbb{Q}$ is the fixed field.

b. We see that $\sigma_3 = \sigma_1\sigma_2$ leaves $\sqrt{6}$ fixed, because $\sqrt{6} = \sqrt{2}\sqrt{3}$ and σ_3 acts on this product by changing the sign of both factors. Thus the fixed field is $\mathbb{Q}(\sqrt{6})$.

c. Because $\sigma_3(\sqrt{2}) = -\sqrt{2}, \sigma_2(\sqrt{3}) = -\sqrt{3}$, and $\sigma_2(\sqrt{6}) = -\sqrt{6}$, the only elements of $\mathbb{Q}(\sqrt{2},\sqrt{3})$ left fixed by both σ_3 and σ_2 are those in $\mathbb{Q}$, so $\mathbb{Q}$ is the fixed field.

16. Because τ_3 moves only $\sqrt{3}$, the subfield of $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5})$ left fixed by τ_3 is $\mathbb{Q}(\sqrt{2},\sqrt{5})$.

17. Because $\tau_3{}^2 = \iota$, the identity, the entire field $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5})$ is left fixed by $\tau_3{}^2$.

18. Because $\tau_2(\sqrt{2}) = -\sqrt{2}$ and $\tau_3(\sqrt{3}) = -\sqrt{3}$, we see that the subfield of $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5})$ left fixed by $\{\tau_2,\tau_3\}$ is $\mathbb{Q}(\sqrt{5})$.

19. Because $\tau_2(\sqrt{2}) = -\sqrt{2}$ and $\tau_5(\sqrt{5}) = -\sqrt{5}$, we see that $\tau_5\tau_2$ leaves both $\sqrt{3}$ and $\sqrt{10}$ fixed, so the subfield of $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5})$ left fixed by $\tau_5\tau_2$ is $\mathbb{Q}(\sqrt{3},\sqrt{10})$.

20. We see that $\tau_5\tau_3\tau_2$ leaves $\sqrt{15}, \sqrt{6}$, and $\sqrt{10}$ fixed. Because $\sqrt{15} = \sqrt{6}\sqrt{10}/2$, we see that we can describe the fixed field of $\tau_5\tau_3\tau_2$ as $\mathbb{Q}(\sqrt{6},\sqrt{10})$.

21. Because every product of one, two or three distinct factors formed from $\sqrt{2}, \sqrt{3}$, and $\sqrt{5}$ is moved by one of τ_2, τ_3, or τ_5, we see that the subfield of $\mathbb{Q}(\sqrt{2},\sqrt{3},\sqrt{5})$ left fixed by $\{\tau_2,\tau_3,\tau_5\}$ is $\mathbb{Q}$.

22. **a.** Because $\tau_2(\sqrt{2}) = -\sqrt{2}$, we see that $\tau_2{}^2(\sqrt{2}) = \tau_2(\tau_2(\sqrt{2})) = \tau_2(-\sqrt{2}) = \sqrt{2}$. Because τ_2 moves neither $\sqrt{3}$ nor $\sqrt{5}$, we see that $\tau_2{}^2 = \iota$, the identity map. Thus τ_2 is of order 2 in $G(E/\mathbb{Q})$. Clearly the same argument shows that both τ_3 and τ_5 are of order 2 in $G(E/\mathbb{Q})$ also.

b. $H = \{\iota, \tau_2, \tau_3, \tau_5, \tau_2\tau_3, \tau_2\tau_5, \tau_3\tau_5, \tau_2\tau_3\tau_5\}$.

	ι	τ_2	τ_3	τ_5	$\tau_2\tau_3$	$\tau_2\tau_5$	$\tau_3\tau_5$	$\tau_2\tau_3\tau_5$
ι	ι	τ_2	τ_3	τ_5	$\tau_2\tau_3$	$\tau_2\tau_5$	$\tau_3\tau_5$	$\tau_2\tau_3\tau_5$
τ_2	τ_2	ι	$\tau_2\tau_3$	$\tau_2\tau_5$	τ_3	τ_5	$\tau_2\tau_3\tau_5$	$\tau_3\tau_5$
τ_3	τ_3	$\tau_2\tau_3$	ι	$\tau_3\tau_5$	τ_2	$\tau_2\tau_3\tau_5$	τ_5	$\tau_2\tau_5$
τ_5	τ_5	$\tau_2\tau_5$	$\tau_3\tau_5$	ι	$\tau_2\tau_3\tau_5$	τ_2	τ_3	$\tau_2\tau_3$
$\tau_2\tau_3$	$\tau_2\tau_3$	τ_3	τ_2	$\tau_2\tau_3\tau_5$	ι	$\tau_3\tau_5$	$\tau_2\tau_5$	τ_5
$\tau_2\tau_5$	$\tau_2\tau_5$	τ_5	$\tau_2\tau_3\tau_5$	τ_2	$\tau_3\tau_5$	ι	$\tau_2\tau_3$	τ_3
$\tau_3\tau_5$	$\tau_3\tau_5$	$\tau_2\tau_3\tau_5$	τ_5	τ_3	$\tau_2\tau_5$	$\tau_2\tau_3$	ι	τ_2
$\tau_2\tau_3\tau_5$	$\tau_2\tau_3\tau_5$	$\tau_3\tau_5$	$\tau_2\tau_5$	$\tau_2\tau_3$	τ_5	τ_3	τ_2	ι

c. An automorphism in $G(E/\mathbb{Q})$ is completely determined by its values on $\sqrt{2}, \sqrt{3}$, and $\sqrt{5}$. Each of these is either left alone or mapped into its negative. Thus there are two possibilities for the value of $\sigma \in G(E/\mathbb{Q})$ on $\sqrt{2}$, two possibilities for $\sigma(\sqrt{3})$, and two possibilities for $\sigma(\sqrt{5})$, giving a total of $2 \cdot 2 \cdot 2 = 8$ automorphisms in all. Because $|H| = 8$, we see that $H = G(E/\mathbb{Q})$.

23. The definition is incorrect. Insert "irreducible" before "polynomial".

Two elements, α and β, of an algebraic extension E of a field F are **conjuate over** F if and only if they are both zeros of the same irreducible polynomial $f(x)$ in $F[x]$.

24. The definition is correct.

25. a. We have $\beta = 3 - \sqrt{2}$ is a conjugate of $\alpha = 3 + \sqrt{2}$. They are both zeros of the polynomial $(x-3)^2 - 2 = x^2 - 6x + 7$ which is irreducible over $\mathbb{Q}$.

b. Because

$$\begin{aligned}\psi_{\alpha,\beta}(\sqrt{2}) &= \psi_{\alpha,\beta}(-3 + (3 + \sqrt{2})) \\ &= \psi_{\alpha,\beta}(-3) + \psi_{\alpha,\beta}(3 + \sqrt{2}) \\ &= -3 + (3 - \sqrt{2}) = -\sqrt{2} = \psi_{\sqrt{2},-\sqrt{2}}(\sqrt{2}),\end{aligned}$$

we see that $\psi_{\alpha,\beta}$ and $\psi_{\sqrt{2},-\sqrt{2}}$ are the same map.

26. We have

$$\sigma_2(0) = 0^2 = 0, \;\; \sigma_2(1) = 1^2 = 1, \;\; \sigma_2(\alpha) = \alpha^2 = \alpha + 1,$$

and

$$\sigma_2(\alpha + 1) = (\alpha + 1)^2 = \alpha^2 + 2 \cdot \alpha + 1 = \alpha^2 + 1 = \alpha + 1 + 1 = \alpha.$$

Thus $\mathbb{Z}_2(\alpha)_{\{\sigma_2\}} = \{0, 1\} = \mathbb{Z}_2$.

27. Using the table for this field in this manual, we find that

$$\sigma_3(0) = 0^3 = 0, \;\; \sigma_3(1) = 1^3 = 1, \;\; \sigma_3(2) = 2^3 = 2,$$

$$\sigma_3(\alpha) = \alpha^3 = 2\alpha, \;\; \sigma_3(2\alpha) = (2\alpha)^3 = \alpha,$$

$$\sigma_3(1 + \alpha) = (1 + \alpha)^3 = 1 + 2\alpha,$$

$$\sigma_3(1 + 2\alpha) = (1 + 2\alpha)^3 = 1 + \alpha,$$

$$\sigma_3(2 + \alpha) = (2 + \alpha)^3 = 2 + 2\alpha,$$

and

$$\sigma_3(2 + 2\alpha) = (2 + 2\alpha)^3 = 2 + \alpha.$$

Thus $\mathbb{Z}_3(\alpha)_{\{\sigma_3\}} = \mathbb{Z}_3$.

28. The map $\sigma_2 : \mathbb{Z}_2(x) \to \mathbb{Z}_2(x)$, where x is an indeterminate, is not an automorphism because the image is $\mathbb{Z}_2(x^2)$. Thus σ is not onto $\mathbb{Z}_2(x)$, but rather maps $\mathbb{Z}_2(x)$ one to one onto a proper subfield of itself.

29. F F T T F T T T T T

30. If α and β are conjugate, then they have the same irreducible polynomial $p(x)$ over F, so both $F(\alpha)$ and $F(\beta)$ are isomorphic to $F[x]/\langle p(x)\rangle$.

31. If $\psi_{\alpha,\beta}$ is an isomorphism of $F(\alpha)$ onto $F(\beta)$, then for every polynomial $f(x) \in F[x]$, we have $f(\alpha) = 0$ if and only if $f(\beta) = 0$, so the monic irreducible polyomial for α over F is the same as the one for β over F.

32. By Corollary 48.5, such an isomorphism must map α onto one of its conjugates over F. Because $\deg(\alpha, F) = n$ there are at most n conjugates of α in $\overline{F}$, for a polynomial of degree n has at most n zeros in a field. On the other hand, Corollary 48.5 asserts that there is exactly one such isomorphism for each conjugate of α over F, so the number of such isomorphisms is equal to the number of conjugates of α over F, which is $\leq n$.

33. We proceed by induction on n. For $n = 1$, Corollary 48.5 shows that σ is completely determined by $\sigma(\alpha_1)$, which must be a conjugate of α_1 over F. Suppose that the statement is true for $n < k$, and let $n = k$. Suppose that σ is known on $\alpha_1, \alpha_2, \cdots, \alpha_{k-1}, \alpha_k$. Let $r = \deg(\alpha_k, F(\alpha_1, \cdots, \alpha_{k-1}))$. Then each element β in $F(\alpha_1, \cdots, \alpha_{k-1}, \alpha_k)$ can be written uniquely in the form

$$\beta = \gamma_0 + \gamma_1\alpha_k + \gamma_2\alpha_k^2 + \cdots + \gamma_{r-1}\alpha_k^{r-1}$$

where $\gamma_i \in F(\alpha_1, \cdots, \alpha_{k-1})$ for $i = 0, 1, \cdots, r-1$ according to Theorem 30.23. By our induction assumption, we know $\sigma(\gamma_i)$ for $i = 0, 1, \cdots, r-1$, and we are assuming that we also know $\sigma(\alpha_k)$. The expression for β and the fact that σ is an automorphism shows that we know $\sigma(\beta)$. This completes our proof by induction.

34. By Corollary 48.5, σ maps each zero of $\text{irr}(\alpha, F)$ onto a zero of this same polyomial. Because σ is an automorphism, it is a one-to-one map of E onto E. By counting, it must map the set of zeros in E of this polynomial onto itself, so it is a permutation of this set.

35. Because $S \subseteq H$, it is clear that $E_H \subseteq E_S$. Let $\alpha \in E_S$, so that $\sigma_i(\alpha) = \alpha$ for all $i \in I$. Then $\sigma_i^{-1}(\alpha) = \alpha$ also. It follows at once that $\sigma_i^n(\alpha) = \alpha$ for all $i \in I$ and all $n \in \mathbb{Z}$. Theorem 7.6 shows that every element of H is a product of a finite number of such powers of the σ_i. Because products of automorphisms are computed by function composition, it follows that α is left fixed by each element in H. Therefore $\alpha \in E_H$, so $E_S \subseteq E_H$ and therefore $E_S = E_H$.

36. **a.** Suppose that $\zeta^i = \zeta^j$ for $i < j \leq p-1$. Then $\zeta^{j-i} = 1$ and ζ would be a zero of $x^{j-i} - 1$ which is of degree less than $p-1$, contradicting the fact that $\Phi_p(x)$ is irreducible. Thus these powers ζ^i for $1 \leq i \leq p-1$ are distinct. Because $\zeta^i \neq 1$ for $1 \leq i \leq p-1$ but $(\zeta^i)^p = (\zeta^p)^i = 1^i = 1$, we see that ζ^i is a zero of $x^p - 1$ that is different from 1 for $1 \leq i \leq p-1$, so these distinct powers of ζ must account for all $p-1$ zeros of $\Phi_p(x)$.

b. Let $\sigma, \tau \in G(\mathbb{Q}(\zeta)/\mathbb{Q})$. Suppose that $\sigma(\zeta) = \zeta^i$ and that $\tau(\zeta) = \zeta^j$. Then

$$\begin{aligned}(\sigma\tau)(\zeta) &= \sigma(\tau(\zeta)) = \sigma(\zeta^j) = [\sigma(\zeta)]^j = (\zeta^i)^j \\ &= \zeta^{ij} = \zeta^{ji} = (\zeta^j)^i = [\tau(\zeta)]^i \\ &= \tau(\zeta^i) = \tau(\sigma(\zeta)) = (\tau\sigma)(\zeta).\end{aligned}$$

Because $(\sigma\tau)(\zeta) = (\tau\sigma)(\zeta)$, Corollary 48.5 shows that $\sigma\tau = \tau\sigma$. Thus $G(\mathbb{Q}(\zeta)/\mathbb{Q})$ is abelian.

c. We know that $B = \{1, \zeta, \zeta^2, \cdots, \zeta^{p-2}\}$ is a basis for $\mathbb{Q}(\zeta)$ over $\mathbb{Q}$. Let $\beta \in \mathbb{Q}(\zeta)$. We can write

$$\frac{\beta}{\zeta} = a_0 + a_1\zeta + a_2\zeta^2 + \cdots + a_{p-2}\zeta^{p-2}$$

for $a_i \in \mathbb{Q}$ for $0 \leq i \leq p-2$. Multiplying by ζ, we see that

$$\beta = a_0\zeta + a_1\zeta^2 + a_2\zeta^3 + \cdots + a_{p-2}\zeta^{p-1} \tag{1}$$

so these powers of ζ do span $\mathbb{Q}(\zeta)$. They are linearly independent because a linear combination of them equal to zero yields a linear combination of the elements in B equal to zero upon division by ζ. Thus the set $\{\zeta, \zeta^2, \cdots, \zeta^{p-1}\}$ is a basis for $\mathbb{Q}(\zeta)$ over $\mathbb{Q}$.

By Theorem 48.3, and Part(**a**), there exist automorphisms σ_i for $i = 1, 2, \cdots, p-1$ in $G(\mathbb{Q}(\zeta)/\mathbb{Q})$ such that $\sigma_i(\zeta) = \zeta^i$. Thus if β in Eq.(1) is left fixed by all such σ_i, we must have $a_0 = a_1 = \cdots = a_{p-2}$ so $\beta = a_0(\zeta + \zeta^2 + \cdots + \zeta^{p-1}) = -a_0$ because ζ is a zero of $\Phi_p(x)$. Thus the elements of $\mathbb{Q}(\zeta)$ left fixed lie in $\mathbb{Q}$, so $\mathbb{Q}$ is the fixed field of $G(\mathbb{Q}(\zeta)/\mathbb{Q})$.

37. Yes, if α and β are transcendentals over F, then $\phi : F(\alpha) \to F(\beta)$, where $\phi(a) = a$ for each $a \in F$ and $\phi\left(\frac{f(\alpha)}{g(\alpha)}\right) = \frac{f(\beta)}{g(\beta)}$ for $f(x), g(x) \in F[x]$ and $g(x) \neq 0$, is an isomorphism. Both $F(\alpha)$ and $F(\beta)$ are isomorphic to $F(x)$ as we saw in Case II under the heading **Simple Extensions** in Section 29.

38. In the notation of Exercise 37, taking $\alpha = x$, we must find all β transcendental over F such that $F(x) = F(\beta)$. This means that not only must β be a quotient of polynomials in x that does not lie in the field F, but also, we must be able to solve and express x as a quotient of polynomials in β. This is only possible if β is a quotient $\frac{ax+b}{cx+d}$ of linear polynomials in $F[x]$, and for this quotient not to be in F, we must require that $ad - bc \neq 0$, so $\frac{a}{c} \neq \frac{b}{d}$.

39. **a.** Let σ be an automorphism of E and let $\alpha \in E$. Then $\sigma(\alpha^2) = \sigma(\alpha\alpha) = \sigma(\alpha)\sigma(\alpha) = \sigma(\alpha)^2$, so σ indeed carries squares into squares.

b. Because the positive numbers in $\mathbb{R}$ are precisely the squares in $\mathbb{R}$, this follows at once from Part(**a**).

c. From $a < b$, we deduce that $b - a > 0$. By Part(**b**), we see that $\sigma(b-a) = \sigma(b) - \sigma(a) > 0$, so $\sigma(a) < \sigma(b)$.

d. Let σ be an automorphism of $\mathbb{R}$. Let $a \in \mathbb{R}$, and find sequences $\{r_i\}$ and $\{s_i\}$ of rational numbers, both converging to a, and satisfying

$$r_i < r_{i+1} < a < s_{i+1} < s_i$$

for all $i \in \mathbb{Z}^+$. By Part(**c**), we see that

$$\sigma(r_i) < \sigma(r_{i+1}) < \sigma(a) < \sigma(s_{i+1}) < \sigma(s_i). \tag{2}$$

The automorphism σ of $\mathbb{R}$ must leave the prime field $\mathbb{Q}$ fixed, because $\sigma(1) = 1$. Thus the inequality (2) becomes

$$r_i < r_{i+1} < \sigma(a) < s_{i+1} < s_i$$

for all $i \in \mathbb{Z}^+$. Because the sequences $\{r_i\}$ and $\{s_i\}$ converge to a, we see that $\sigma(a) = a$, so σ is the identity automorphism.